# Final Project Report: Electric Vehicle Market Performance & Insights

A Comprehensive Business Intelligence & Analytical Framework for Global EV Market Trends

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## 1. Introduction

### 1.1 Project Overview

This project involves designing and developing an interactive Business Intelligence (BI) dashboard using Tableau to analyze the global Electric Vehicle (EV) market. It visualizes key metrics such as total sales, battery chemistry efficiency, pricing trends, and geographical production footprints.

### 1.2 Purpose

The purpose is to provide automotive strategy executives and market analysts with a centralized, data-driven tool to make informed decisions regarding supply chain investments, battery platform development, and regional market targeting.

## 2. Ideation Phase

### 2.1 Problem Statement

Executives and supply chain managers struggle with scattered, static reports, making it difficult to correlate EV battery chemistries with market pricing, range, and global sales. This fragmentation risks poor strategic platform decisions.

### 2.2 Empathy Map Canvas

- **Target User:** Automotive Strategy Executive / EV Market Analyst.
- **Pains:** Frustrated by scattered data, fear of investing massive capital into the wrong battery platform, and unseen supply chain risks.
- **Gains:** Needs a unified, interactive Tableau dashboard for instant insights and visual proof of price vs. range to confidently present recommendations.

### 2.3 Brainstorming

The solution conceptualized was a 3x3 interactive dashboard accompanied by a 6-point Tableau Story, driven by a raw CSV/Excel dataset. The focus was on regional adoption, battery types (NMC, LFP, Sodium-Ion), and segment pricing.

## 3. Requirement Analysis

### 3.1 Customer Journey Map

The customer journey involves three stages: (1) Access & Discovery, opening the Tableau Public link; (2) Exploring & Filtering, drilling down by region or battery type; (3) Extracting Insights, navigating the Story tabs and exporting reports for executive presentations.

## 3.2 Solution Requirements

**Functional:** Data ingestion, cleaning, KPI calculation, and dynamic dashboard generation with multi-faceted filtering.

**Non-Functional:** Usability optimized via Tableau typography, sub-5-second render latency, public data security, and 24/7 web availability.

## 3.3 Data Flow Diagram

Raw Data (CSV) → Tableau Data Connection (Cleaning/Formatting) → Tableau Local Extract (.twbx) → Tableau Desktop (Visualization) → Tableau Public Server (Hosting) → End-User Web Dashboard

## 3.4 Technology Stack

| Component | Selected Technology | Purpose |
|---|---|---|
| BI Tool | Tableau Desktop Public Edition | Visual interface and dashboard construction |
| Hosting | Tableau Public | Cloud distribution and interactive web access |
| Data Source | CSV / Excel Datasets | Baseline operational and market data |
| Logic Engine | Tableau Calculated Fields | Dynamic metric and KPI derivation |

# 4. Project Design

## 4.1 Problem Solution Fit

The fragmented data problem is solved by consolidating global EV market data into a single, interactive 9-visual Tableau Dashboard that visually maps sales, pricing, and battery chemistries geographically.

## 4.2 Proposed Solution

To ingest raw EV datasets into Tableau, clean the data using calculated fields, and build a cohesive analytical experience featuring real-time KPI scorecards, interactive filtering, and geographical production mapping.

## 4.3 Solution Architecture

Data flows from raw CSV/Excel files into Tableau Desktop for ingestion and transformation, is stored in Tableau Data Extracts (.twbx), rendered as a 3x3 dashboard, and published to the Tableau Public Cloud Server for end-user interaction via a web browser.

# 5. Project Planning & Scheduling

Executed over 3 sprints. Sprint 1 focused on data access and KPI calculation. Sprint 2 focused on geospatial mapping and interactive filtering. Sprint 3 focused on business insights storytelling and Tableau Public deployment.

# 6. Functional and Performance Testing

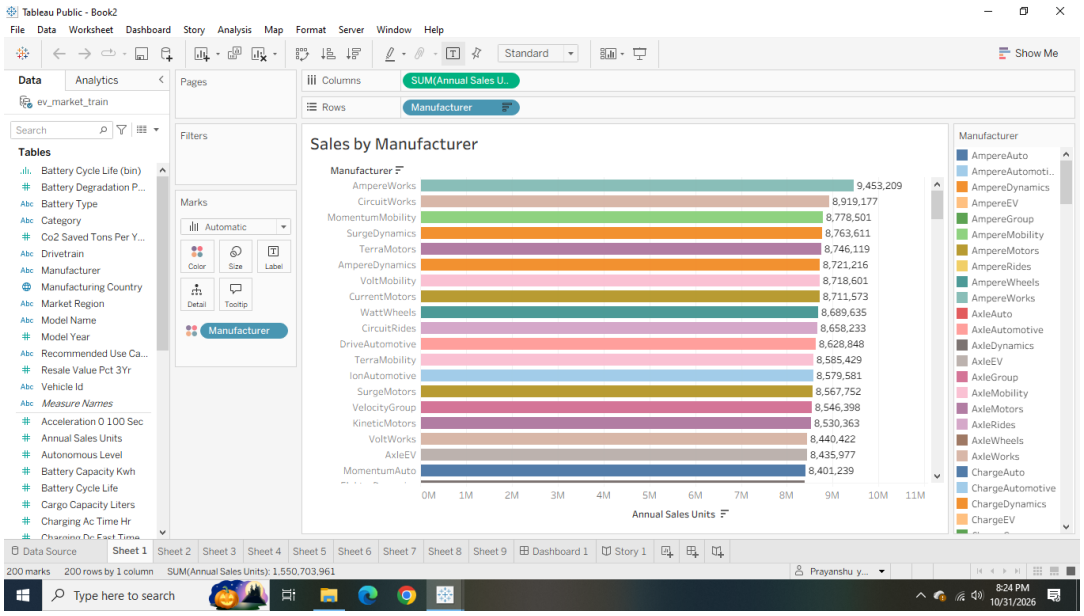
## 6.1 Performance Testing

- **Data Rendered:** Global EV market data loaded successfully for 1,550.7M EV sales.
- **Filters:** Category, Battery Type, and Manufacturer filters tested and working seamlessly.

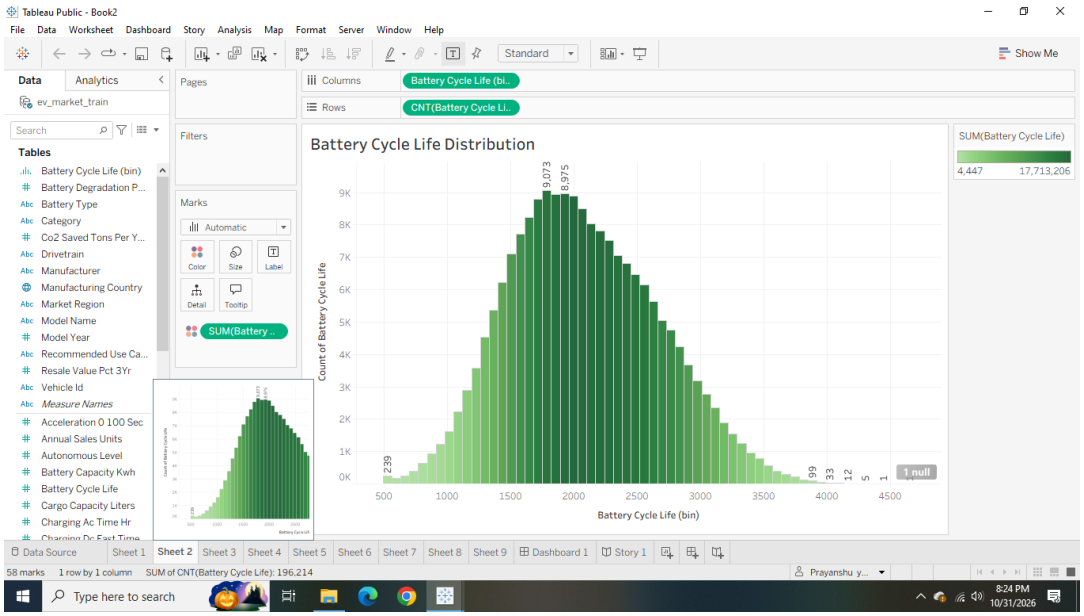
- **Calculations:** Total EV Sales, Average Price, and Median Range KPIs validated for accuracy.
- **Visualizations:** 9 dashboard visuals and 6 Story points successfully rendered.

# 7. Results

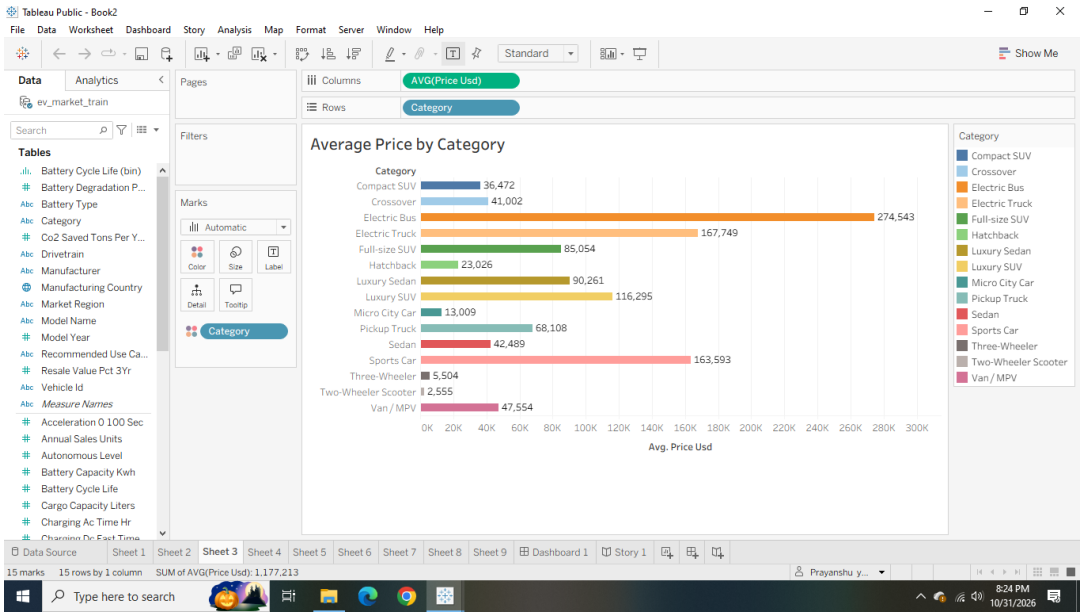
## 7.1 Output Screenshots



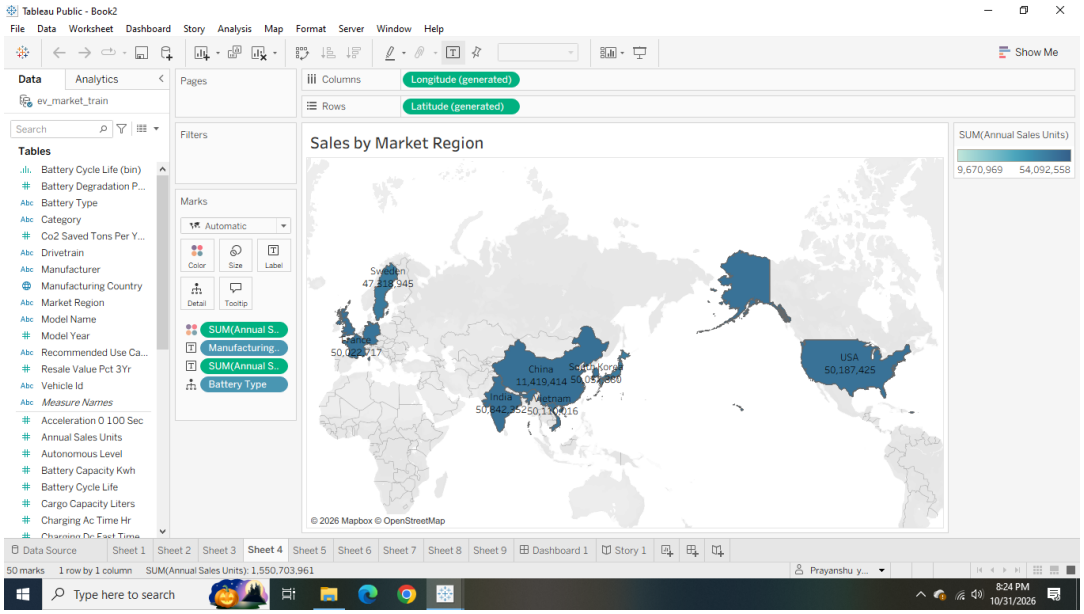

Sales by Manufacturer (Tableau worksheet)



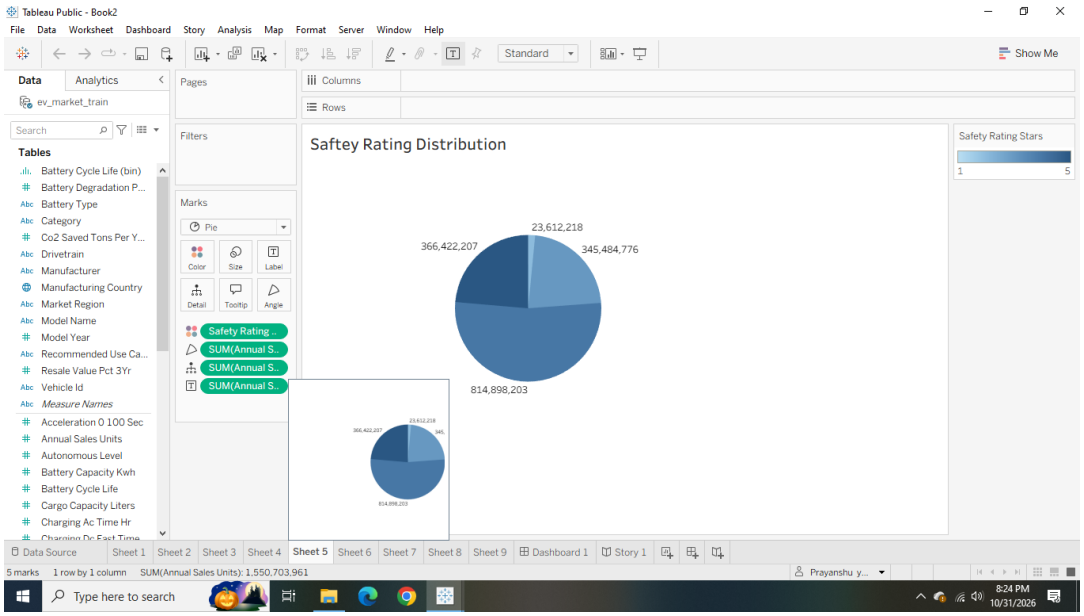

Battery Cycle Life Distribution (Tableau worksheet)



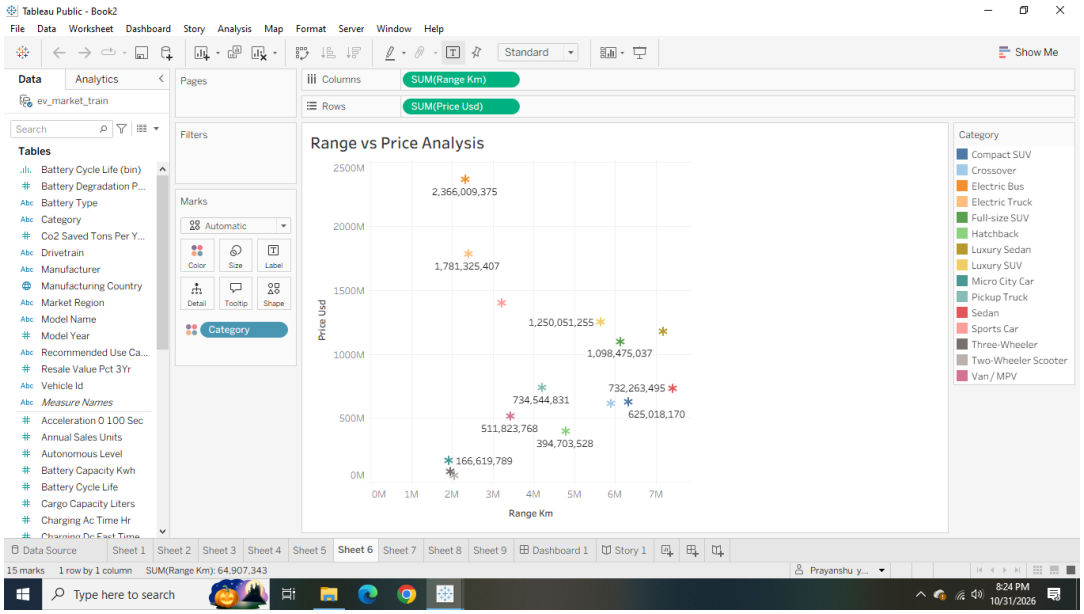

Average Price by Category (Tableau worksheet)



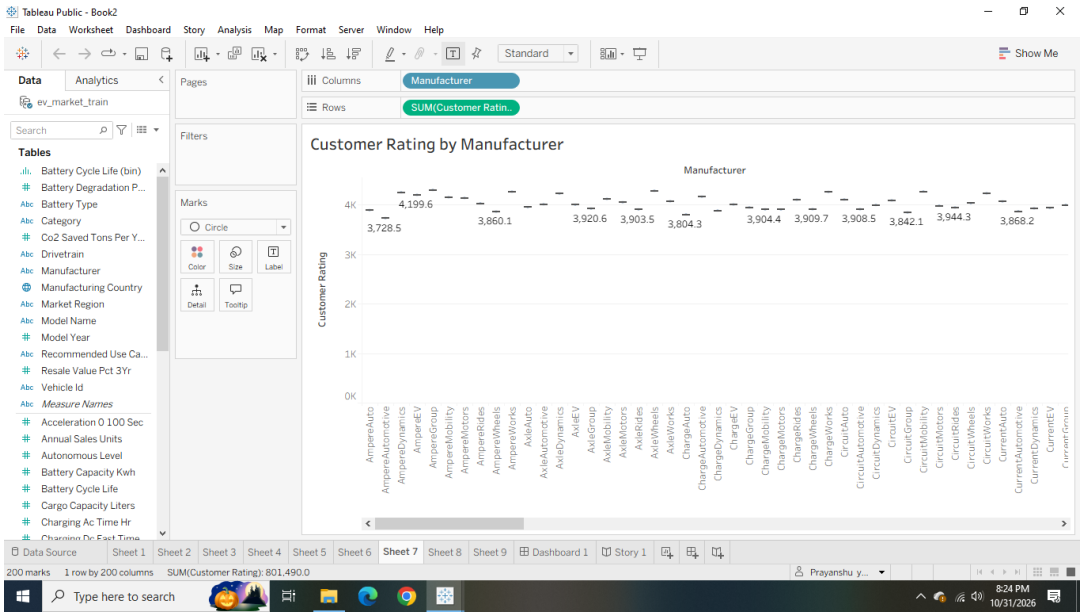

Sales by Market Region (Tableau worksheet)



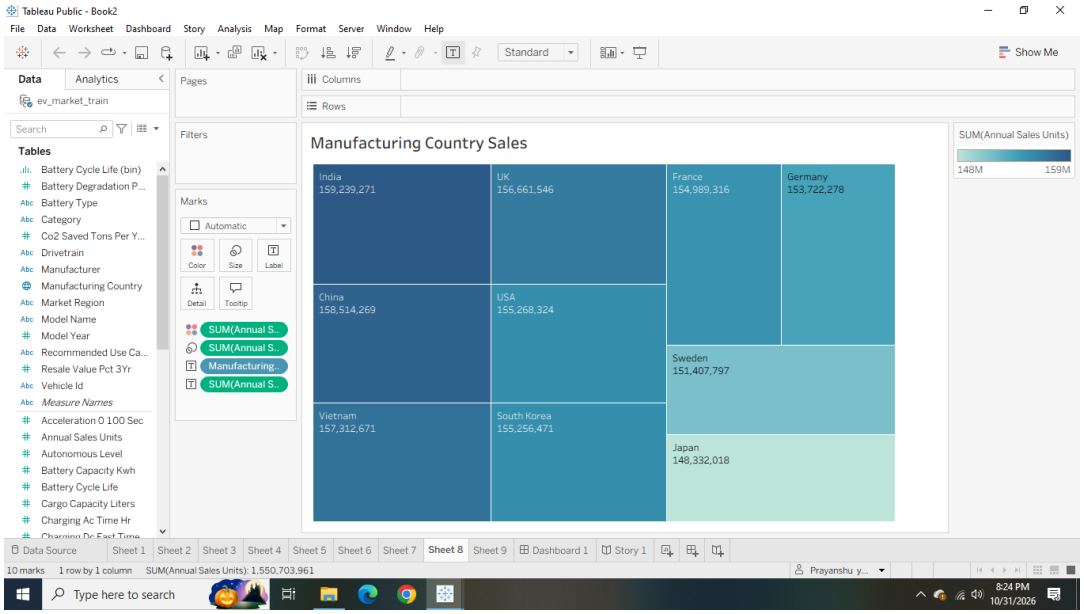

Safety Rating Distribution (Tableau worksheet)



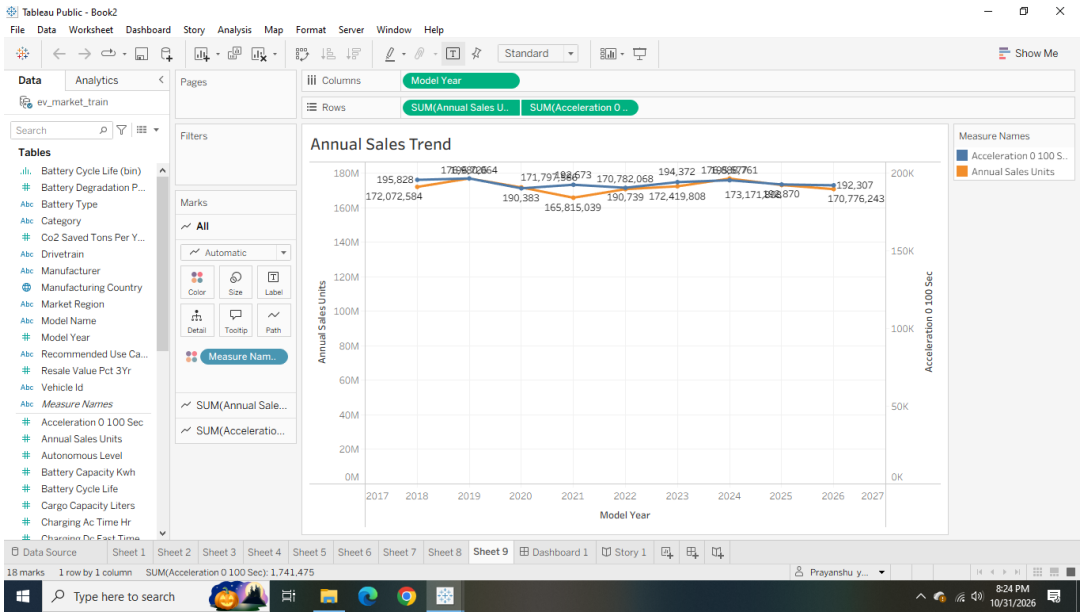

Range vs Price Analysis (Tableau worksheet)




Customer Rating by Manufacturer (Tableau worksheet)




Manufacturing Country Sales (Tableau worksheet)




Annual Sales Trend (Tableau worksheet)

## 8. Advantages and Disadvantages

**Advantages:** Provides instant, interactive insights across multiple metrics. Web-accessible via Tableau Public without requiring local software installation. Easily readable executive summary through the 6-point Data Story.

**Disadvantages:** Relies on a static CSV file; requires manual updates for new data. Tableau Public hosts data publicly, so confidential corporate data cannot be used.

## 9. Conclusion

The Electric Vehicle Market Performance and Insights project bridges the gap between raw, fragmented automotive data and actionable business intelligence. By leveraging Tableau, the dashboard provides stakeholders with clear correlations between battery technology, pricing, and regional production, supporting data-driven strategic decisions.

## 10. Future Scope

Integrate a live database connection for automated, real-time data updates. Implement predictive analytics to forecast future EV sales based on historical trends. Expand the dataset to include detailed charging infrastructure availability by region.

## 11. Appendix

Dataset Link: Dataset Folder (Google Drive)

Story Link: Interactive Story (Tableau Public)

Dashboard Link: Interactive Dashboard (Tableau Public)

Video Link: Project Demo Video (Google Drive)

Templates: Templates Folder (Google Drive)